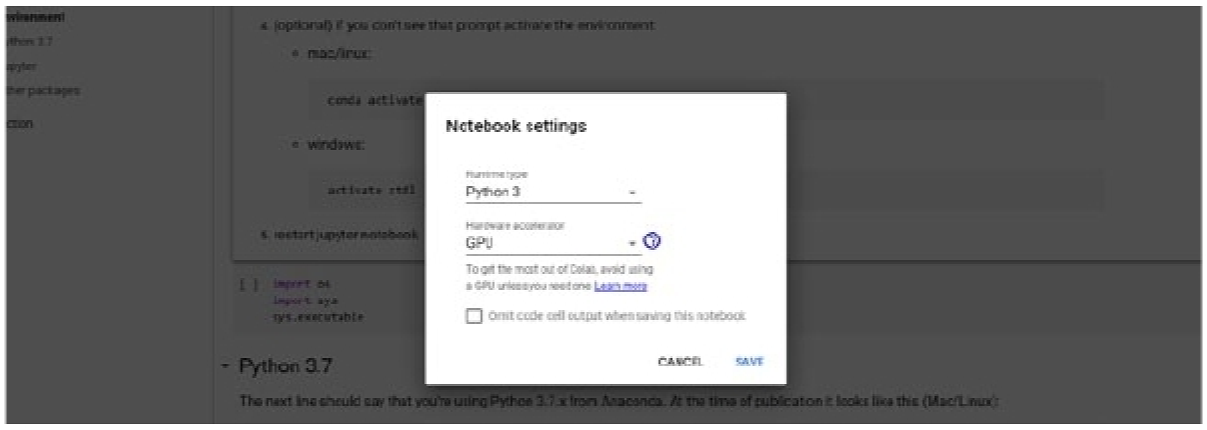

Figure 7: Google Colab GPU settings

Figure 8: Google Colab Google Drive mount

and are purely CPUs. So, what is the recourse? The answer is: Google Colab. It's an attempt from Google to provide on-demand VMs with GPUs to run your tests. You can potentially mount your Google Drive as well to get the data from it. The steps given below will help you.

1. Log in to *colab.research.google.com.*
2. Look at *File* in the top menu.
3. You can write a new notebook or you can import an existing one.
4. You can go to the *Edit > Notebook* settings, where you can choose the hardware accelerator as the GPU. Refer to Figure 7.
5. You can even mount your Google Drive using the following code. Refer to Figure 8.

```
from google.colab import drive
drive.mount('/content/drive')
import tensorflow as tf

print (tf.test.gpu_device_name())
```

Deep learning has been evolving at a tremendous pace since the last couple of years. It is necessary to keep technically abreast with this emerging green field to stay relevant in the technical landscape. This article describes the taxonomy and various tools available at our disposal. While some are proprietary/commercial, there are plenty of FOSS tools around also. One of them is Keras. Familiarity with basic Python scripting is good enough to start with Keras. It abstracts the low-level languages (e.g., CUDA) and platforms (e.g., TensorFlow) beautifully to get a kickstart in this paradigm. It's easy to install even on your laptops, and on Linux or Windows. A sample program of the MNIST data set digit detection has also been presented to give a glimpse of the anatomy of a DL program. This is just the beginning, and the road ahead is endless and enriching.

Note: The MNIST digit detection example mentioned in this article can be found in the following GitHub link: *https://github.com/pradip-interra/MNIST-Digit-Detection*

END

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